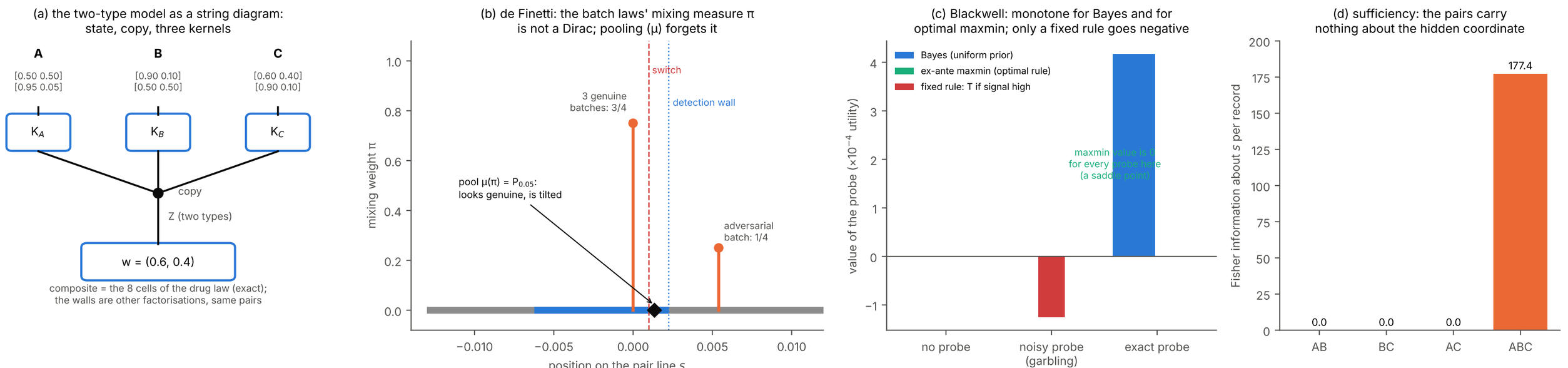
(a) the two-type model as a string diagram:
state, copy, three kernels
A
B
C
[0.50 0.50]
[0.95 0.05]
[0.90 0.10]
[0.50 0.50]
[0.60 0.40]
[0.90 0.10]
K_A
K_B
K_C
copy
Z (two types)
w = (0.6, 0.4)
composite = the 8 cells of the drug law (exact);
the walls are other factorisations, same pairs
(b) de Finetti: the batch laws' mixing measure π
is not a Dirac; pooling (μ) forgets it
mixing weight π
1.0
0.8
0.6
0.4
0.2
0.0
switch
detection wall
3 genuine
batches: 3/4
pool μ(π) = $P_{0.05}$:
looks genuine, is tilted
adversarial
batch: 1/4
−0.010
−0.005
0.000
0.005
0.010
position on the pair line s
(c) Blackwell: monotone for Bayes and for
optimal maxmin; only a fixed rule goes negative
Bayes (uniform prior)
ex-ante maxmin (optimal rule)
fixed rule: T if signal high
value of the probe (×10^{-4} utility)
4
3
2
1
0
−1
maxmin value is 0
for every probe here
(a saddle point)
no probe
noisy probe
(garbling)
exact probe
(d) sufficiency: the pairs carry
nothing about the hidden coordinate
Fisher information about s per record
200
175
150
125
100
75
50
25
0
177.4
0.0
0.0
0.0
AB
BC
AC
ABC